

PART-III : REAL ANALYSIS

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SET

Set: It is well defined collection of object e.g. $N, I, I^+, I^-, Q, R, \dots$

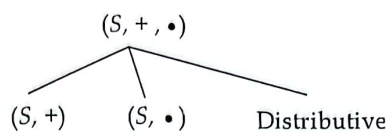
Equality of sets: When sets have exactly the same elements e.g. $\{a, b, c\} = \{b, a, c\}$

- Universal set
- Difference set
- Complement of a set - A^c

Equivalent set: Two sets are said to be equivalent ($A \sim B$) if there exists a one-one correspondence between their elements.



Field structure:

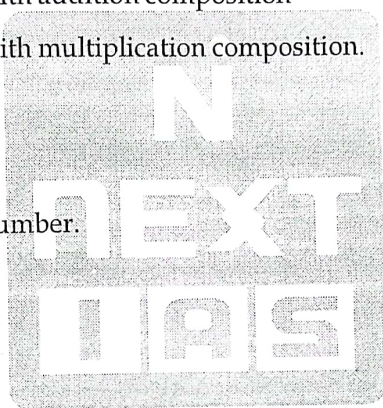


Order structure: A field S is an ordered field if it satisfied following properties:

1. Law of trichotomy
2. Transitivity
3. Compatibility of order relation with addition composition
4. Compatibility of order relation with multiplication composition.

Examples: $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields
 $\mathbb{N}$ & $\mathbb{I}$ are not fields

Q. Show that $\sqrt{8}$ is not a rational number.



INTERVALS

Open and closed: A subset A of $\mathbb{R}$ is an interval if A contains atleast two distinct elements and every element lies between any two members of A .

- Open interval (a, b) or $] a, b [$
- Closed interval $[a, b]$
- Semi-closed or semi-open $] a, b]$

Bounded and Unbounded sets: A subset S of real numbers is said to be bounded above if $\exists$ a real number k such that every member of S is less than or equal to k , i.e.

$$x \leq k \quad \forall x \in S$$

The number k is called upper bound of S . If no such number k exists, the set is said to be unbounded above.

Bounded below: If $\exists$ a real number k s.t. every number of S is greater than or equal to k , i.e.

$$k \leq x \quad \forall x \in S$$

The number k is called a lower bound of S . If no such number exists, the set is said to be unbounded below.

Bounded set: (i) bounded below (ii) bounded above

Least upper bounded (l.u.b.): If the set of all upper bounds of a set S has the smallest member say M , then M is called the *least upper bounded (l.u.b.)* or *supremum* of S .

- M may / may not exist
- M may / may not belong to S .

Alternative way of describing M : M satisfies two properties

- $x \leq M, \quad \forall x \in S$
- For any +ve number ϵ , $\exists$ a number $y \in S$ s.t. $y > M - \epsilon$

Greatest lower bounded (g.l.b.): If the set of all lower bounds of a set S has the greatest member, say m , then ' m ' is called the *greatest lower bounded (g.l.b.)* or the *infimum* of ' S '. If may or may not exist, it may or may not belong to S .

Alternative way of describing ' m ':

- m is the lower bound of S i.e. $x \geq m, \quad \forall x \in S$
- For any +ve number ϵ , $\exists$ a number $z \in S$, such that $z < m + \epsilon$

ORDER COMPLETENESS IN $\mathbb{R}$

Every non-empty set of real numbers which is bounded above has the supremum in $\mathbb{R}$

or

Every non-empty set of real numbers which is bounded below has the infimum (or glb) in $\mathbb{R}$.

Q. The set of rational number is not order complete.



Archimedean property of real numbers:

If a and b are any two positive real numbers then there exists a positive integer ' n ' such that $na > b$.

Corollary: For any $\epsilon > 0$, $\exists$ a +ve n (integer) s.t. $\frac{1}{n} < \epsilon$



Neighbourhood of a point: $N(N \subseteq R)$ is called the nbd of point a if there exist an open interval I containing a and contained in N i.e. $a \in I \subseteq N$.

Examples:

1. The set R of real numbers is the nbd of each of its points.
 2. The set Q of rationals is not the nbd of any of its point.
 3. The open interval $]a, b[$ is nbd of each of its points.
 4. The closed interval $[a, b]$ is nbd of each points of (a, b) but is not a nbd of the end point.
 5. The null set ϕ is a nbd of each of its points in the sense that there is no point in ϕ of which it is not a nbd.
- Q.** A non-empty finite set is not a nbd of any point.



Note: Superset of a nbd of a point x is also a nbd of x .

Interior point of a set: A point x is an interior point of a set S if S is a nbd of x .

Note: (i) The set of all interior points of a set is called the interior of the set.

(ii) Denoted by S^i or $\text{int } S$.

Example: Interior of set N or I or Q is the null set, but interior of R is R .

OPEN SET

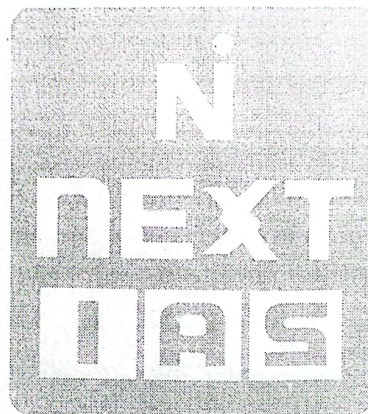
A set S is said to be open if it is nbd of its points.

Every point of an open set is an interior point i.e. $S' = S$

Example:

- $\mathbb{R}$: open set
- $\mathbb{Q}$: not open
- $[a, b]$: not open
- ϕ : open
- Finite set : not open
- Set $\left\{\frac{1}{n} : n \in \mathbb{N}\right\}$ is not open

Example: Show that every open interval is an open set.



Theorem: The intersection of any finite number of open sets is open.

LIMIT POINT OF A SET

A real number ξ is a limit point of a set $S (\subset \mathbb{R})$ if every nbd of ξ contains infinite number of members of S .

- ξ is a limit point of a set S if for any nbd. N of ξ , $N \cap S$ is an infinite set.
- A limit point is also called a cluster point, a condensation point or an accumulation point.
- It may or may not be a member of the set.
- A finite set cannot have a limit point
- It is not necessary that an infinite set must possess a limit point.
- A set may have no limit point, a unique limit point, a finite or an infinite number of limit points.

(The idea is that points of S different from ξ get arbitrarily close to ξ or pile up at ξ)

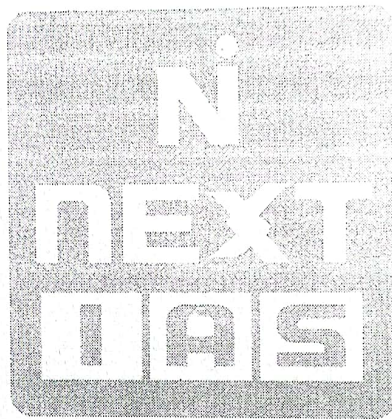
Derived set (S'): The set of all limit points of a set S is called the derived set of S , denoted by S' .

Example:

1. The set I has no limit point. Derived set $I' = \phi$
2. Every point of $\mathbb{R}$ is a limit point $\mathbb{R}' = \mathbb{R}$
3. $\mathbb{Q}' = \mathbb{R}$
4. The set $\left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$ has only one limit point 0.
5. Every point of the closed set $[a, b]$ is its limit point
 $[a, b]' = [a, b]$
6. Every point of the open interval $]a, b[$ is its limit point a, b are also limit points
 Therefore, $]a, b[' = [a, b]$

BOLZANO-WEIERSTRAUSS THEOREM (for set)

"Every infinite bounded set has a limit point."



Closed set: If each of its limit points is a member of the set

$$\boxed{S' \subseteq S}$$

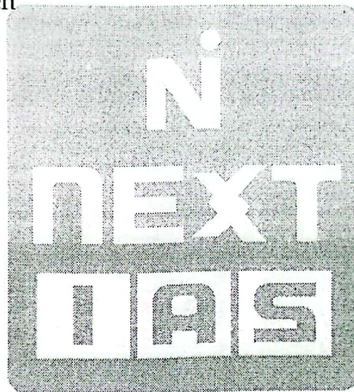
Note:

1. Concept of closed and open sets are neither mutually exclusive nor exhaustive.
2. Not closed is not equivalent to open e.g. $]a, b]$

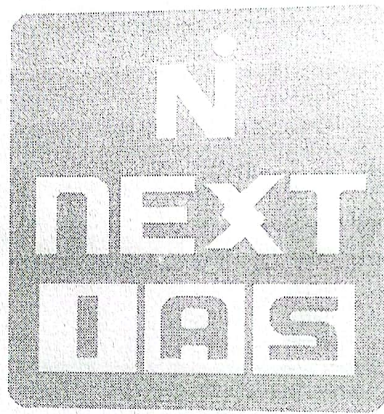
Example:

1. $[a, b]$: closed, not open
2. Set $[0, 1] \cup [2, 3]$ which is not interval is closed
3. $\mathbb{R}, \emptyset$: closed, open
4. $\mathbb{Q}$: not closed not open
5. Every finite set is closed set

A set is closed iff its complement is open



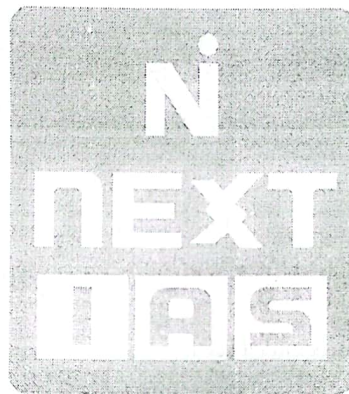
The union of two closed set is a closed set.



COUNTABLE AND UNCOUNTABLE SETS

A set which is either empty, finite or *countably infinite* is called countable otherwise *uncountable*.

Countably infinite: if the set is equivalent to the set N of natural numbers.



REAL SEQUENCES

A function whose domain is the set N of natural numbers and range, a set of real numbers is called a real sequence.

Since, the domain is always N , a sequence is specified by the values $S_n, n \in N$

Thus, a sequence may be denoted as

$$\{S_n\}, n \in N \text{ or } \{S_1, S_2, S_3, \dots\}$$

$S_1, S_2, S_3, \dots$ are the first, second, third, terms of sequence. The m^{th} and n^{th} terms are treated as distinct terms even if they have same value. The number of terms in a sequence is always infinite.

Example: $\{S_n\} = \{(-1)^n\}, n \in N$

$$\{S_n\} = \left\{\frac{1}{n}\right\}, n \in N$$

Range: Set consisting of all distinct elements of a sequence

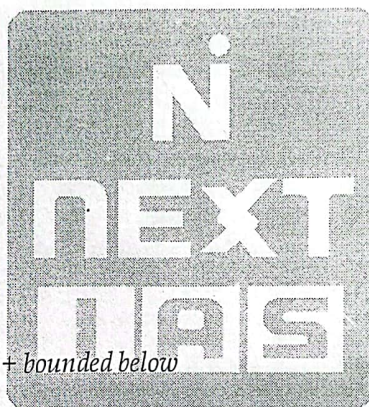
e.g. $\{S_n\} = \{(-1)^n\}$

Range: $\{1, -1\}$

Bounds of sequence:

- Bounded above: $S_n \leq k \forall n \in N$
- Bounded below: $S_n \geq k \forall n \in N$

Bounded sequence = bounded above + bounded below



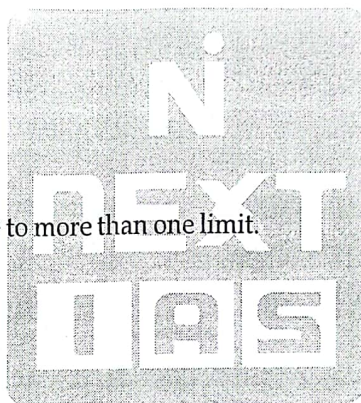
Convergence of a sequence: A sequence $\{S_n\}$ is said to converge to a real number l if for such $\epsilon > 0$, there exists a positive integer m (depending on ϵ) such that $|S_n - l| < \epsilon$, for $\forall n \geq m$

Expressed as: $S_n \rightarrow l$ as $n \rightarrow \infty$ or $\lim_{n \rightarrow \infty} S_n = l$

Note: Since, $l - \epsilon < S_n < l + \epsilon$ for all $n \geq m$, therefore $S_n < l + \epsilon$ for infinite number of terms

Theorem: Every convergent sequence is bounded.

Theorem: A sequence cannot converge to more than one limit.



Limit points for a sequence: A real number ξ is said to be a limit point of a sequence $\{S_n\}$ if every neighbourhood of ξ contains an infinite number of members of the sequence i.e. $|S_n - \xi| < \epsilon$ for infinitely many values of n .

A limit point is also called a cluster point or a point of condensation.

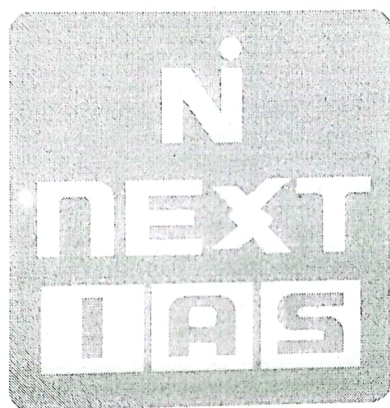
Example:

1. The constant sequence $\{S_n\}$ where $S_n = 1, \forall n \in N$ has the only limit point 1. Range = $\{1\}$.
2. $\{S_n\}$ where $S_n = (-1)^n$. Limit points = 1, -1.



BOLZANO-WEIERSTRAUSS THEOREM (for sequences)

"Every bounded sequence has a limit point."



CONVERGENT SEQUENCES

A sequence may have no limit point, a unique limit point or a finite or an infinite number of limit points.

Bounded sequence with unique limit point is called convergent sequence.

A necessary and sufficient condition for a sequence $\{S_n\}$ to converge to l is that to each $\epsilon > 0$, there corresponds a positive integer m such that $|S_n - l| < \epsilon; \forall n \geq m$

Non convergent sequences:

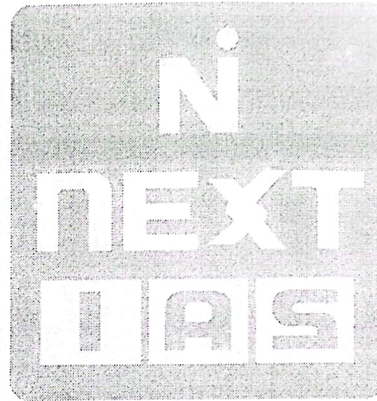
(i) Bounded sequences $\Rightarrow$ oscillate finitely

e.g. $\{1 + (-1)^n\}; \{(-1)^n \left(1 + \frac{1}{n}\right)\}$

(ii) Unbounded sequence

e.g. $\{n^2\}; \left\{m + \frac{1}{n}\right\} m, n \in N$

Q. Show that $\lim_{n \rightarrow \infty} \frac{3 + 2\sqrt{n}}{\sqrt{n}} = 2$



Q. $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$

CAUCHY'S GENERAL PRINCIPLE OF CONVERGENCE

A necessary and sufficient condition for the convergence of a sequence $\{S_n\}$ is that for each $\epsilon > 0$ there exists a positive integer m such that

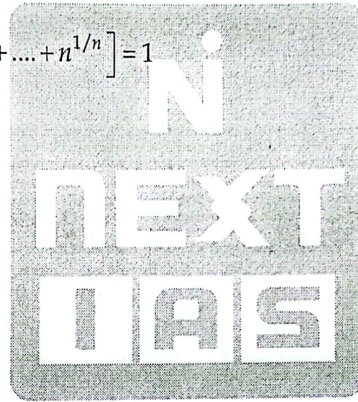
$$|S_{n+p} - S_n| < \epsilon, \forall n \geq m, p \geq 1$$

Q. Show that the sequence $\{S_n\}$, where $S_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$ cannot converge.

SOME IMPORTANT THEOREMS

1. If $\lim a_n = a$ and $a_n \geq 0 \forall n$, then $a \geq 0$
 2. If $\{a_n\}, \{b_n\}$ be two sequences such that
 - (i) $a_n \leq b_n \forall n$
 - (ii) $\lim a_n = a, \lim b_n = b$, then $\boxed{a \leq b}$
 3. Sandwich theorem: If $\{a_n\}, \{b_n\}, \{c_n\}$ are three sequences such that
 - (i) $a_n \leq b_n \leq c_n \forall n$
 - (ii) $\lim a_n = \lim c_n = l$ then $\lim b_n = l$
 4. If $\lim_{n \rightarrow \infty} a_n = l$, then $\lim_{n \rightarrow \infty} \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right) = l$ } Cauchy's 1st theorem on limits
 5. If $\lim_{n \rightarrow \infty} a_n = l$, then $\lim_{n \rightarrow \infty} (a_1, a_2, \dots, a_n)^{1/n} = l$
 6. If $\lim a_n = a$ and $\lim b_n = b$, then $\lim_{n \rightarrow \infty} \left(\frac{a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1}{n} \right) = ab$ } Cesaro's theorem
 7. If $\lim_{n \rightarrow \infty} \left(\frac{U_{n+1}}{U_n} \right) = a$ then $\lim_{n \rightarrow \infty} (U_n)^{1/n} = a \ (U_n > 0)$,
 i.e. $\lim (U_n)^{1/n} = \lim \frac{U_{n+1}}{U_n}$ } it should exist $\rightarrow$ (Cauchy's second theorem on limit)
 8. If $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = l$, and $|l| < 1$, then $\lim a_n = 0$
 9. $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = l > 1$, then $\lim a_n = \infty$
- Q. Show that $\{b_n\}$ where $b_n = \left[\frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{(2n)^2} \right]$ converges to 0.

Q. Show that $\lim_{n \rightarrow \infty} \frac{1}{n} [1 + 2^{1/2} + 3^{1/3} + \dots + n^{1/n}] = 1$

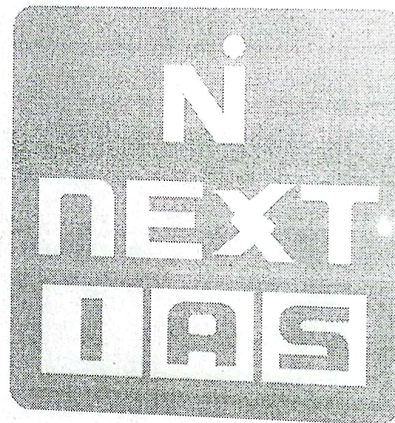


Q. $\lim (n)^{1/n} = 1$



Q. Show that the sequence $\{a_n^{1/n}\}$ and $\{b_n^{1/n}\}$ where (i) $a_n = \frac{3n}{(\underline{n})^3}$ and (ii) $b_n = \frac{n^n}{(n+1)(n+2)\dots(n+n)}$

Q. Show that for any real number x , $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$.

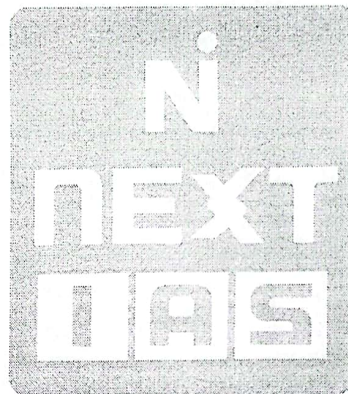


Monotonic sequences:

- (i) Monotonic increasing: $S_{n+1} \geq S_n$
- (ii) Monotonic decreasing: $S_{n+1} \leq S_n$ (strictly - no equality)

Thus, monotonic sequence cannot oscillate. They either converge or diverge.

Theorem: A necessary and sufficient condition for the convergence of a monotonic sequence is that it is bounded.



Subsequences: If $\{S_n\} = \{S_1, S_2, \dots\}$ be a sequence, then any infinite succession of its terms, picked out in any way (but preserving the original order) is called a subsequence of $\{S_n\}$.

Example:

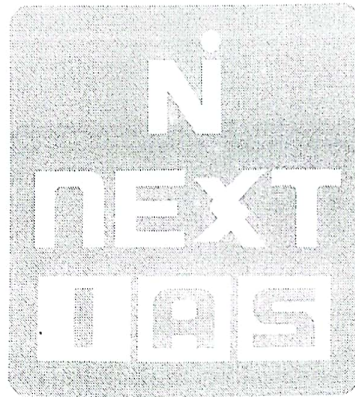
- 1. $\{S_2, S_4, S_6, \dots, S_{2n}, \dots\}$ is a subsequence of $\{S_n\}$
- 2. $\{S_7, S_8, S_9, \dots\}$ is a subsequence of $\{S_n\}$

Note: A sequence $\{S_n\}$ converge to 'S' iff its every subsequence converges to 'S'.

Q. Show that the sequence $\{S_n\}$ where $S_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n}$, $\forall n \in \mathbb{N}$ is convergent.



Q. Show that the sequence $\{a_n\}$ defined by $a_{n+1} = \frac{1}{2} \left(a_n + \frac{9}{a_n} \right)$, $n \geq 1$ and $a_1 > 0$ converges to 3.



INFINITE SERIES (POSITIVE TERMS)

Let $\langle U_n \rangle$ be a given sequence, then a symbol of the form $u_1 + u_2 + \dots + u_n + \dots$ is called an infinite series.

Notation: $\sum_{n=1}^{\infty} U_n$ or simply ΣU_n .

The sequence $\langle S_n \rangle$ is the sequence of partial sums of the series ΣU_n

i.e. $S_n = U_1 + U_2 + \dots + U_n$

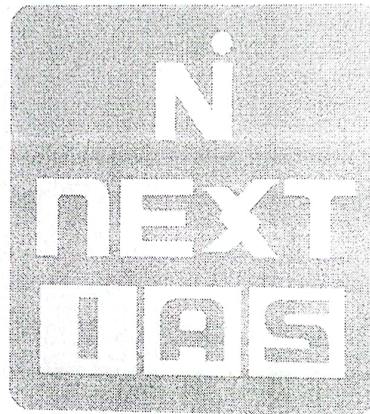
The infinite series ΣU_n is said to be convergent if the sequence $\langle S_n \rangle$ of its partial sums is convergent.

If $\lim_{n \rightarrow \infty} S_n = S$, then S is called the sum of the series ΣU_n and we write $S = \Sigma U_n$

Note: 1. The replacement, addition or omission of finite no. of terms of a series ΣU_n has no effect on its convergence.

2. The convergence of series remains unchanged if each of its term is multiplied by a non-zero constant.

Q. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

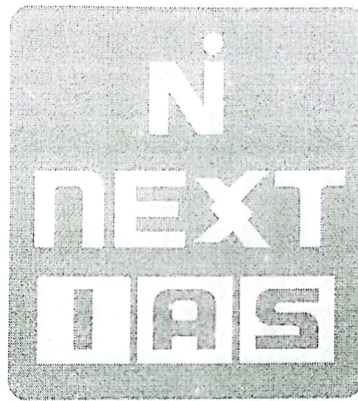


NECESSARY AND SUFFICIENT CONDITION

Cauchy's general principle of convergence for series: A necessary and sufficient condition for a series $\sum U_n$ to converge is that for each $\epsilon > 0$ there exist a positive integer m such that

$$|U_{m+1} + U_{m+2} + \dots + U_n| < \epsilon \quad \forall n > m$$

Q. Show that the series $\sum \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \dots$ does not converge.



OTHER THEOREMS

1. If the series $\sum U_n$ converge, then $\lim_{n \rightarrow \infty} U_n = 0$
2. A NSC for the convergence of a positive term series $\sum_{n=1}^{\infty} U_n$ is that the sequence $\langle S_n \rangle$ of the partial sums of the series defined by $S_n = U_1 + U_2 + \dots + U_n$ is bounded above.
3. If each term of a series $\sum U_n$ of positive terms, does not exceed the corresponding term of a convergent series $\sum V_n$ of +ve terms, then $\sum U_n$ is convergent.
4. If $\lim_{n \rightarrow \infty} U_n \neq 0$, $\sum U_n$ diverges.
5. If $\sum U_n$ and $\sum V_n$ are two convergent series, then $\sum (U_n + V_n)$ and $\sum (U_n - V_n)$ are also convergent.

STANDARD RESULT

1. The positive term infinite geometrical series $1 + r + r^2 + \dots + r^n + \dots$ is convergent iff $r < 1$.
2. The positive term series $\sum \frac{1}{n^p}$ is convergent iff $p > 1$.

VARIOUS TESTS FOR CONVERGENCE OF SERIES

1. **Comparison Tests (First Type):** If $\sum U_n$ and $\sum V_n$ are two positive term series and $k \neq 0$ (a fixed positive number independent of n) and there exist a positive integer m such that $U_n \leq kV_n, \forall n \geq m$, then

(i) $\sum U_n$ is convergent, if $\sum V_n$ is convergent

(ii) $\sum V_n$ is divergent, if $\sum U_n$ is divergent.

Limit form: $\lim_{n \rightarrow \infty} \left(\frac{U_n}{V_n} \right) = l$, where l is non zero finite no; then the two series converge or diverge together.

2. **Comparison Test (Second Type):** $\exists$ a positive integer m such that $\frac{U_n}{U_{n+1}} \geq \frac{V_n}{V_{n+1}}, \forall n \geq m$ then

(i) $\sum U_n$ is convergent if $\sum V_n$ is convergent.

(ii) $\sum V_n$ is divergent if $\sum U_n$ is divergent.

3. **D'Alembert's Ratio Test:**

$$\lim_{n \rightarrow \infty} \frac{U_n}{U_{n+1}} = l \quad \begin{array}{ll} l > 1 & \text{Converge} \\ l < 1 & \text{Diverge} \\ l = 1 & \text{Test fail} \end{array}$$

4. **RAABE's Test:**

$$\lim_{n \rightarrow \infty} n \left(\frac{U_n}{U_{n+1}} - 1 \right) = l \quad \begin{array}{ll} l > 1 & \text{Converge} \\ l < 1 & \text{Diverge} \\ l = 1 & \text{Test fail} \end{array}$$

5. **Logarithmic Test:**

$$\lim_{n \rightarrow \infty} \left(n \log \frac{U_n}{U_{n+1}} \right) = l \quad \begin{array}{ll} l > 1 & \text{Converge} \\ l < 1 & \text{Diverge} \\ l = 1 & \text{Test fail} \end{array}$$

6. **Cauchy's Root Test:**

$$\lim_{n \rightarrow \infty} (U_n)^{1/n} = l \quad \begin{array}{ll} l < 1 & \text{Converge} \\ l > 1 & \text{Diverge} \\ l = 1 & \text{Test fail} \end{array}$$

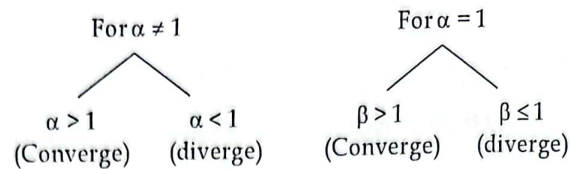
7. **Integral Test:** Let's say U is a non-negative monotonic decreasing integrable function s.t. $U(n) = U_n$, then the

series $\sum_{n=1}^{\infty} U_n$ and $\int_1^{\infty} U(x) dx$ cvg or dvg together.

8. **Gauss Test:**

$$\frac{U_n}{U_{n+1}} = \alpha + \frac{\beta}{n} + \frac{\gamma_n}{n^p}$$

where $\alpha > 0$, $p > 1$ and $\{\gamma_n\}$ is a bounded sequence, then



9. De Morgan's and Bertrand's Test:

$$\log \left[\left\{ n \left(\frac{U_n}{U_{n+1}} - 1 \right) - 1 \right\} \log n \right] = l \quad \begin{array}{ll} l > 1 & \text{Converge} \\ l < 1 & \text{Diverge} \\ l = 1 & \text{Test fail} \end{array}$$

10. Second Log Ratio Test:

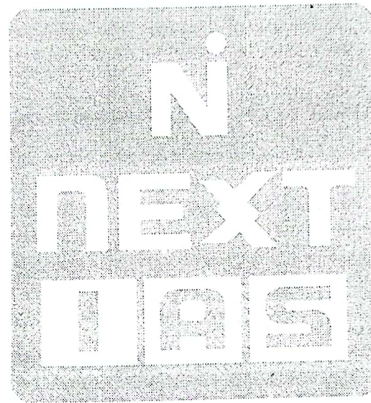
$$\lim_{n \rightarrow \infty} \left\{ \left(n \log \frac{U_n}{U_{n+1}} - 1 \right) \log n \right\} = l \quad \begin{array}{ll} l > 1 & \text{Converge} \\ l < 1 & \text{Diverge} \\ l = 1 & \text{Test fail} \end{array}$$

11. Cauchy's Condensation Test: Let $U(n)$ be positive monotonically decreasing function then $\sum U(n)$ and $\sum a^n U(a^n)$ converge or diverge together ('a' is +ve integer ≥ 2).

Q Examine the convergence of the series: $\sum \left\{ (n^3 + 1)^{1/3} - n \right\}$

Q. $\sum_{n=2}^{\infty} \frac{1}{n^2 \log n}$

Q. $\sum \frac{n!}{n^n}$

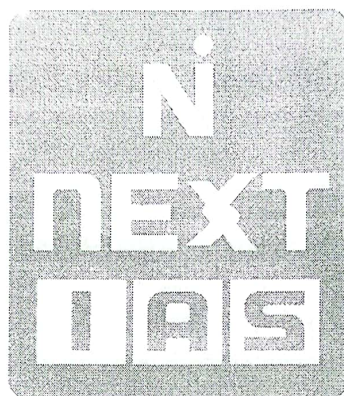


Q. $\left(\frac{2^2}{1^2} - \frac{2}{1}\right)^{-1} + \left(\frac{3^3}{2^3} - \frac{3}{2}\right)^{-2} + \left(\frac{4^4}{3^4} - \frac{4}{3}\right)^{-3} + \dots$

Q. $\sum \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots (2n)} \cdot \frac{x^{2n}}{2n}; x \text{ being non-negative.}$

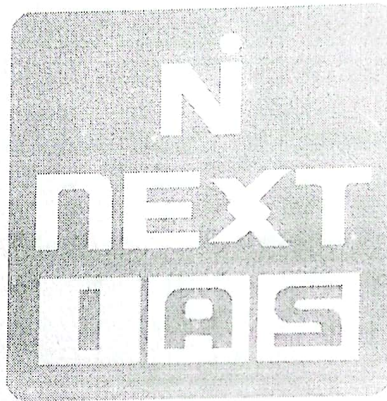


Q. $\sum \frac{1.3.5.....(4n-5)(4n-3)}{2.4.6.....(4n-4)(4n-2)} \cdot \frac{x^{2n}}{4n}; x > 0$



Q. $1 + \frac{x}{1!} + \frac{2^2 x^2}{2!} + \frac{3^2 x^3}{3!} + \dots$, for $x > 0$.

Q. $x + x^{1+\frac{1}{2}} + x^{1+\frac{1}{2}+\frac{1}{3}} + \dots$



Q. $1 + \left(\frac{1}{2}\right)^p + \left(\frac{1.3}{2.4}\right)^p + \left(\frac{1.3.5}{2.4.6}\right)^p + \dots$



Q. Using Cauchy's integral test, show that the series $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ is convergent if $p > 1$ and divergent if $0 < p \leq 1$.

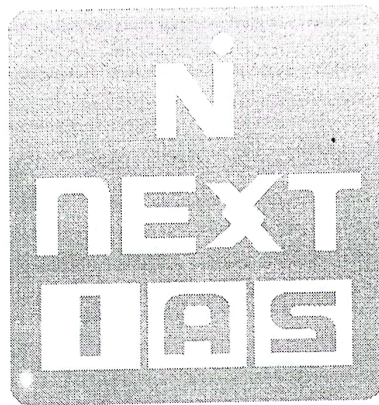
Q. $\frac{(\log 2)^2}{2^2} + \frac{(\log 3)^2}{3^2} + \dots + \frac{(\log n)^2}{n^2} + \dots$



NEXT IAS**INFINITE SERIES WITH POSITIVE AND NEGATIVE TERMS**

Example: $\sum (-1)^n = -1 + 1 - 1 + 1$

This series is not convergent even though the sequence of partial sums is bounded.



ABSOLUTE CONV. AND CONDITIONAL CONV.

A series $\sum U_n$ is said to be *absolutely convergent* if the positive term series $\sum |U_n|$ formed by the moduli of the terms of the series is convergent.

e.g. $\sum U_n = 1 - \frac{1}{2} + \frac{1}{2^2} - \frac{1}{2^3} + \dots$ is absolutely convergent because $\sum |U_n| = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots$ is convergent.

Conditional convergence: Converge without being absolute convergent.

e.g. $\sum U_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is conditionally convergent because $\sum U_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$ is not convergent series.

Note: Every absolutely convergent series is convergent.

ALTERNATING SERIES

Terms are alternately positive and negative.

Form: $U_1 - U_2 + U_3 - U_4 + \dots$, where $U_n > 0 \forall n \in \mathbb{N}$ is an alternating series.

e.g. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

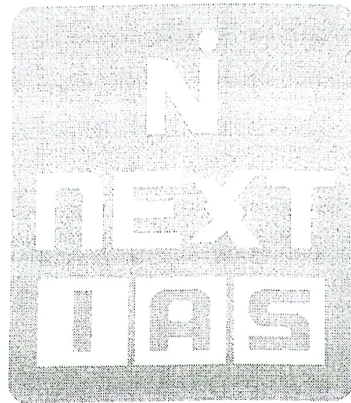
Leibnitz theorem: Let $\langle U_n \rangle$ be sequence s.t.

- (a) $U_n \geq 0$
- (b) $U_{n+1} \leq U_n$
- (c) $\lim_{n \rightarrow \infty} U_n = 0$

Then the alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} U_n$ is convergent.

Q Test for absolute convergence of the series $\sum (-1)^{n-1} \frac{n^2}{(n+1)!}$

Q Test for absolute convergence and convergence $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\log(n+1)}$



DIRICHLET'S THEOREM

Let

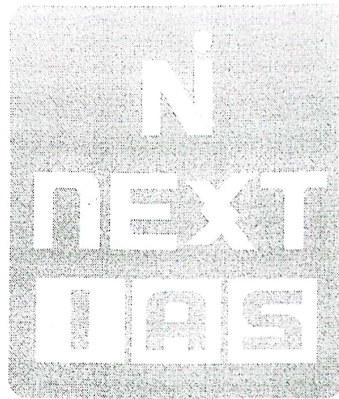
- (a) $\langle U_n \rangle$ be a positive monotonically decreasing sequence with limit 0 and
- (b) $\sum_{n=1}^{\infty} a_n$ be an infinite series such that the sequence of its partial sums is bounded, then $\sum_{n=1}^{\infty} a_n U_n$ is convergent



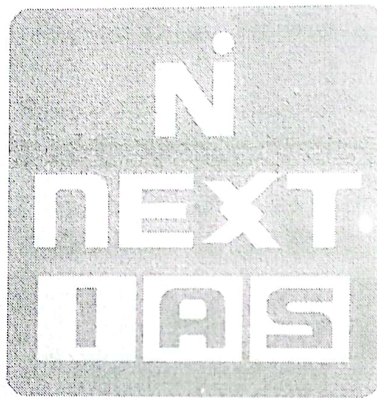
ABEL'S THEOREM

1. If $\langle U_n \rangle$ is a positive monotonically decreasing and
2. $\sum a_n$ is a convergent series, then $\sum a_n U_n$ is convergent.

Q.
$$\sum \frac{(n^3 + 1)^{1/3} - n}{\log n}$$



Q. Show that $1 - \frac{1}{4 \cdot 3} + \frac{1}{4^2 \cdot 5} - \frac{1}{4^3 \cdot 7} + \dots$ is convergent



REARRANGEMENTS OF SERIES

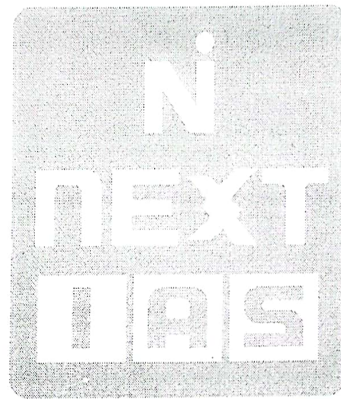
ΣU_n is a rearrangement of a series ΣV_n if every terms of ΣU_n is a term of ΣV_n and vice-versa.

e.g. $1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \dots$ is rearrangement of the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

Note:

1. Every rearrangement of a positive term convergent series is convergent and the sum also does not change.
2. Every rearrangement of an absolute convergent series is convergent and the sum also does not change.
3. Sequence $\langle \gamma_n \rangle$ where $\gamma_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \log n$ is convergent.
4. A conditionally convergent series can be made to converge to any real number, to diverge to $\infty / -\infty$ or oscillate finitely or infinitely.

Q. Let $A : 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \xrightarrow{\text{rearrangement}} B : 1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \dots$



CAUCHY PRODUCT OF TWO INFINITE SERIES

Given: $\sum a_n$ and $\sum b_n$

Take $C_n = a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1$

$$= \sum_{r=1}^n a_r b_{n-r+1}$$

Then $\sum_{n=1}^{\infty} C_n$ is the Cauchy product of two given series

$\sum a_n = \alpha$	$\sum b_n = \beta$	$\sum C_n$
Abs	Abs	Abs, $\alpha\beta$
Abs	Cvg	Cvg $\alpha\beta$
Cvg	Abs	Cvg $\alpha\beta$
Cvg	Cvg	May be $\alpha\beta$ May not be

Q If $a_n = b_n = \frac{(-1)^{n-1}}{\sqrt{n}}$ and $C_n = \sum a_r b_{n-r+1}$, then show that $\sum a_n$ and $\sum b_n$ are conditionally convergent and their Cauchy product $\sum C_n$ is non-convergent.

$\sum |a_n| = \sum \frac{1}{\sqrt{n}}$ is divergent, therefore $\sum a_n$ and $\sum b_n$ are conditionally convergent.

$$\text{Now, } C_n = \sum_{r=1}^n a_r b_{n-r+1} = (-1)^n \left(\frac{1}{\sqrt{n \times 1}} + \frac{1}{\sqrt{2(n-1)}} + \dots + \frac{1}{\sqrt{1 \times n}} \right)$$

$$\Rightarrow |C_n| > \frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2}} + \dots \text{ to } n \text{ times}$$

$$\Rightarrow |C_n| \geq 1 \Rightarrow \sum C_n \text{ is non-convergent.}$$

UNIFORM CONVERGENCE

So far, we have considered almost exclusively sequence and series whose terms were numbers. It was only in particularly simple cases that the terms depended on a variable.

We shall now consider sequence and series, whose terms depend on a variable i.e. whose terms are real valued functions defined on an interval as domain.

Therefore, we denote the terms by $f_n(x)$ and consider sequences and series of the form $\{f_n\}$ and $\sum f_n$ respectively.

Pointwise Convergence

Let f_n be a real valued function defined in a set S for each n .

For each $C \in S$, there corresponds a real number sequence $\{f_n(C)\}$ with values $f_1(C), f_2(C), \dots, f_n(C), \dots$

We suppose that each of the sequences arising for different members of S is convergent.

Thus, we define in a natural way a real valued function say f , with domain S such that its value $f(C)$ for $C \in S$ is $\lim \{f_n(C)\}$.

The function ' f ' is referred to as the pointwise limit of the sequence $\{f_n\}$ of the functions.

Equivalently, we say that a sequence $\{f_n\}$ converges to ' f ' pointwise on the set A if, for each $\epsilon > 0$ and each $x \in S$ there exist a positive integer m depending on ϵ or x s.t. $|f_n(x) - f(x)| < \epsilon$ whenever $n \geq m$.

Example: Show that the sequence (f_n) where $f_n(x) = x^n$, $x \in [0, 1]$ is pointwise convergent. Also find the limit.

Example: Consider the sequence of function $f_n(x) = \cos nx$ for $-\infty < x < \infty$ i.e. $x \in \mathbb{R}$. Show that the sequence is not convergent pointwise for every real x .

Uniform Convergence

A sequence of functions (f_n) converges uniformly on A to a function f if for each $\epsilon > 0$ there is an integer m (independent of x but dependent on ϵ) s.t. for all $x \in [a, b]$, $|f_n(x) - f(x)| < \epsilon, \forall n \geq m$

Note:

1. Uniform convergence $\rightarrow$ pointwise convergence
2. Pointwise convergence: m depends on both x and ϵ

Uniform convergence: m depends on ϵ only. i.e. for each $\epsilon > 0$, it is possible to find one integer m which will do for $\forall x \in [a, b]$

Similarly, for series,

$\sum f_n$ is said to converge uniformly on $[a, b]$ if the sequence (S_n) of its partial sums defined by $S_n(x) = \sum_{i=1}^n f_i(x)$ converges uniformly on $[a, b]$

or, if for $\epsilon > 0$ and for all $x \in [a, b]$ there exist an integer m (independent of x and dependent on ϵ) such that for all x in $[a, b]$

$$|f_1(x) + f_2(x) + \dots + f_n(x) - f(x)| < \epsilon \quad \forall n \geq m$$

Cauchy's criterion for UC

The NSC for a sequence of function $(f_n(x))$ defined on $[a, b]$ to converge uniformly: if for every $\epsilon > 0$ and $\forall x \in [a, b]$, $\exists m$ such that

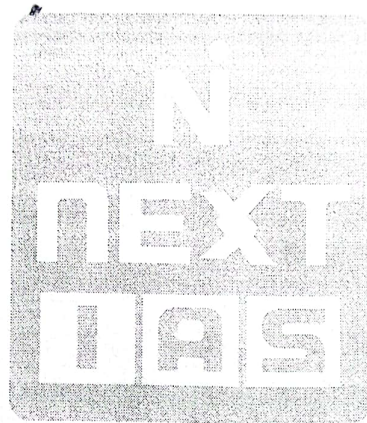
$$|f_{n+p}(x) - f_n(x)| < \epsilon \quad \forall n \geq m, p \geq 1 \quad \text{or} \quad |f_n(x) - f_k(x)| < \epsilon \quad \forall n, k \geq m$$

For series: A series of function $\sum f_n$ defined on $[a, b]$ converges uniformly on $[a, b]$ iff for every $\epsilon > 0$ and $\forall x \in [a, b]$, there exist an integer m such that

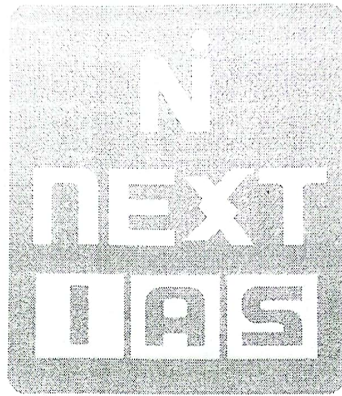
$$|f_{k+1}(x) + f_{k+2}(x) + \dots + f_n(x)| < \epsilon \quad \forall n, k \geq m$$

Significance of the concept of UC

The most important question for a sequence (series) of variable terms, will usually be whether, and to what extent, properties belonging to terms viz., boundedness, continuity, integrability, differentiability etc. are transferred to the limit function of the corresponding sequence (series).



Q Test for uniform convergence, the sequence $(f_n(x))$ where $f_n(x) = \frac{nx}{1+n^2x^2}$, for all real x .



- Q. Show that the sequence $(f_n(x))$ where $f_n(x) = x^n$ is uniformly convergent on $[0, k]$, $k < 1$ and only pointwise convergent $[0, 1]$.



TESTS FOR UNIFORM CONVERGENCE

- (1) For sequence
Let (f_n) be a sequence of functions, s.t.

$$\lim_{n \rightarrow \infty} f_n(x) = f(x), x \in [a, b]$$

and let $M_n = \sup_{x \in [a, b]} |f_n(x) - f(x)|$

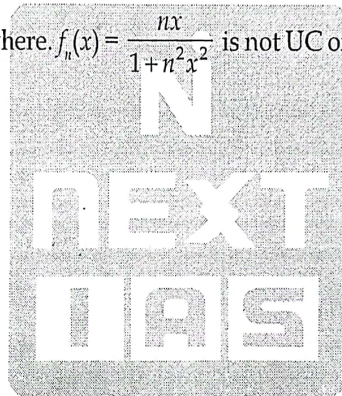
Then, $f_n \rightarrow f$ uniformly on $[a, b]$ iff $M_n \rightarrow 0$ as $n \rightarrow \infty$.

- (2) For series (Weierstrauss M test)

A series of functions $\sum f_n$ will converge uniformly & absolutely on $[a, b]$ if there exists a convergent series

$$\sum M_n \text{ of positive numbers such that for all } x \in [a, b] \quad |f_n(x)| \leq M_n \quad \forall n.$$

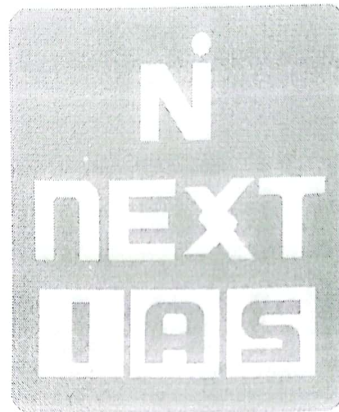
- Q. Show that the sequence $(f_n(x))$, where $f_n(x) = \frac{nx}{1+n^2x^2}$ is not UC on any interval containing zero.



- Q. Show that the sequence $(f_n(x))$ where $f_n(x) = nxe^{-nx^2}$, $x \geq 0$ is not UC on $[0, k]$, $k > 0$.



Q. Show that the series $\sum_{n=1}^{\infty} \frac{x}{n(1+x^2n)}$ converges uniformly.



Other Tests for Series

Since Weierstrass M-tests can be used for only those series which are absolutely cvg hence, we need few more general tests:

(1) **Abel's Test** : Consider series $\sum f_n(x)b_n(x)$ if,

(a) $\sum f_n(x)$ is UC in $[a, b]$

(b) $b_n(x)$ is positive, monotonic decreasing and bounded for all values of n and x concerned, then $\sum f_n(x)b_n(x)$ is also UC.

(2) **Dirichlet's Test** : Consider series $\sum f_n(x)b_n(x)$, if

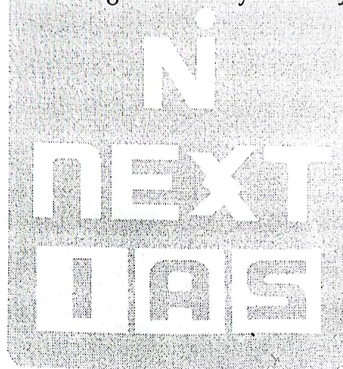
(a) $b_n(x)$ is a monotonic decreasing sequence converging uniformly to zero in $[a, b]$.

(b) $\exists$ a positive real number k s.t.

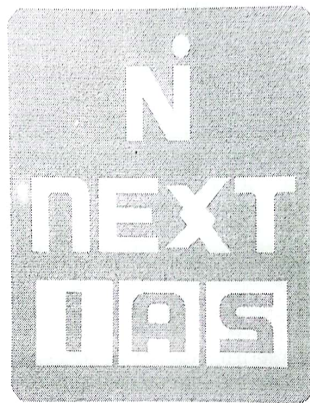
$$\left| \sum_{r=1}^n f_r(x) \right| \leq k \quad \forall n,$$

Then, $\sum f_n(x)b_n(x)$ is UC on $[a, b]$.

Q. Prove that the series $\sum (-1)^n \frac{x^2 + n}{n^2}$, converges uniformly for every bounded interval, but does not converge absolutely for any value of x .



- Q. Prove that the series $\sum \frac{\cos n\theta}{n^p}$ and $\sum \frac{\sin n\theta}{n^p}$ converges uniformly for all values of $p > 0$ in an interval $[\alpha, 2\pi - \alpha]$, where $0 < \alpha < \pi$.



PROPERTIES OF UNIFORMLY CONVERGENT SEQUENCES AND SERIES

We saw earlier that fundamental properties of f_n do not in general hold for pointwise function f . We shall now show that roughly speaking these properties hold for the limit function f when the convergence is uniform.

Theorem 1 :

If a sequence (f_n) converges uniformly in $[a, b]$ and x_0 is a point of $[a, b]$ s.t.

$$\lim_{x \rightarrow x_0} f_n(x) = a_n, n = 1, 2, 3$$

then, (i) (a_n) converges and

$$(ii) \lim_{x \rightarrow x_0} f(x) = \lim_{n \rightarrow \infty} a_n$$

or
$$\left[\lim_{x \rightarrow x_0} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow x_0} f_n(x) \right]$$

Theorem 1B :

If a series $\sum f_n(x)$ converges uniformly to f in $[a, b]$ and x_0 is a point in $[a, b]$ s.t.

$$\lim_{x \rightarrow x_0} f_n(x) = a_n, (n = 1, 2, 3, \dots, n)$$

then (i) $\sum_{n=1}^{\infty} a_n$ converges

and (ii) $\lim_{x \rightarrow x_0} f(x) = \sum_{n=1}^{\infty} a_n$

i.e. $\lim_{x \rightarrow x_0} \sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} \lim_{x \rightarrow x_0} f_n(x).$

{The limit of the sum function of a series = sum of the series of limits of functions}

UNIFORM CONVERGENCE AND CONTINUITY

For sequences

If $(f_n(x))$ is a sequence of continuous functions on an interval $[a, b]$, and if $f_n \rightarrow f$ uniformly on $[a, b]$, then f is continuous on $[a, b]$.

For series

If a series $\sum f_n$ converges uniformly to f in an interval $[a, b]$ and its terms f_n are continuous at a point x_0 of the interval, then the sum function f is also continuous at x_0 .

Note:

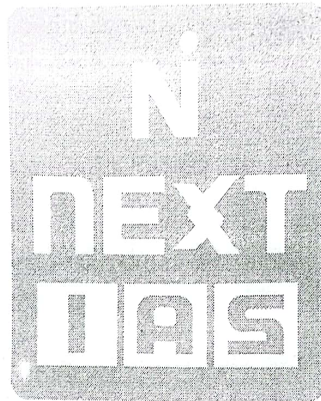
If the sum function (limit function in case of sequence) of continuous terms is not continuous on an interval, the convergence cannot be uniform.

Examples

(1) Sequence (x^n) or $\tan^{-1} nx$ of continuous function has discontinuous limit function on $[0, 1]$. Therefore the convergence is not uniform on $[0, 1]$.

(2) The sum function of the series $\sum (1-x)x^n$ is $f(x) = \begin{cases} 1, & x \neq 1 \\ 0, & x = 1 \end{cases}$ which is discontinuous on $[0, 1]$. Therefore the series is not uniform convergent on $[0, 1]$.

Q. Show that the series $\sum \frac{x}{(nx+1)((n-1)x+1)}$ is UC on any interval, $[a, b]$, $0 < a < b$, but only pointwise on $[0, b]$.



UNIFORM CONVERGENCE AND INTEGRATION

A. Sequence

If a sequence $\{f_n\}$ converges uniformly to f on $[a, b]$, and each function f_n is integrable, then f is integrable on $[a, b]$, and the sequence $\left\{ \int_a^x f_n dt \right\}$ converges uniformly to $\int_a^x f dt$ on $[a, b]$

$$\text{i.e., } \int_a^x f dt = \lim_{n \rightarrow \infty} \int_a^x f_n dt, \forall x \in [a, b]$$

B. Series

If a series $\sum f_n$ converge uniformly to f on $[a, b]$ and each term $f_n(x)$ is integrable, then f is integrable on $[a, b]$

and the series $\int_a^x f_n dt$ converges uniformly to $\int_a^x f dt$ on $[a, b]$,

$$\text{i.e., } \int_a^x f dt = \sum_{n=1}^{\infty} \left(\int_a^x f_n dt \right), \forall x \in [a, b]$$

[We then say that the series is integrable term by term]

Q. Show that the sequence $\{f_n\}$ where

$$f_n(x) = \begin{cases} n^2 x & 0 \leq x \leq 1/n \\ -n^2 x + 2n & 1/n \leq x \leq 2/n \\ 0 & 2/n \leq x \leq 1 \end{cases}$$

Q. $f_n(x) = nx e^{-nx}$, $n = 1, 2, 3$. UC?



UNIFORM CONVERGENCE AND DIFFERENTIATION**(A) Sequences**

Let (f_n) be a sequence of differentiable functions on $[a, b]$ such that it converges at least at one point $x_0 \in [a, b]$. If the sequence of differentials (f'_n) converges uniformly to G on $[a, b]$, then the given sequence (f_n) converges uniformly on $[a, b]$ to f , and $f'(x) = G(x)$.

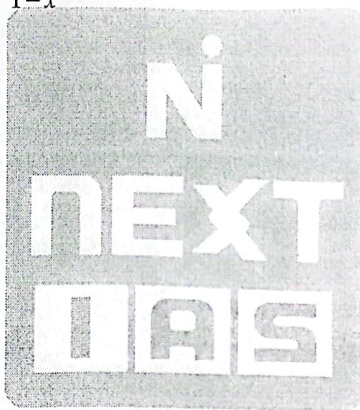
(B) Series

Let $\sum f_n$ be a series of differentiable functions on $[a, b]$ such that it converges at least at one point $x_0 \in [a, b]$.

If the series of differentials $\sum f'_n$ converges uniformly to G on $[a, b]$, then the given series $\sum f_n$ converges uniformly on $[a, b]$ to f , and $f'(x) = G(x)$.

Q. Show that for $-1 < x < 1$

$$\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \frac{8x^7}{1+x^8} + \dots = \frac{1}{1-x}$$



THE RIEMANN INTEGRAL

At elementary stage, the subject of integration is generally introduced as the inverse of differentiation, so that a function F is called an integral of a given function f , if $F'(x) = f(x)$.

Historically speaking the subject arose in connection with the evaluation of areas of plane regions and thus amounted to finding out the limit of a sum when the number of terms tended to infinity, each term tending to zero.

To formulate an independent theory of integration, Riemann gave purely arithmetic treatment to subject and thus developed the subject entirely free from the intuitive dependence on geometric concepts.

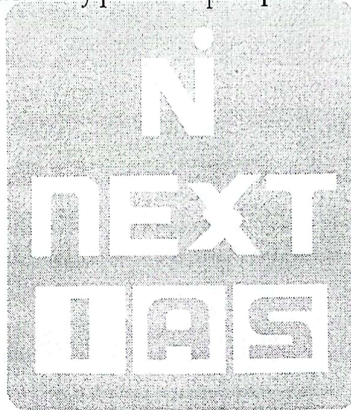
PARTITION: Let $[a, b]$ be a given interval. By a Partition P of $[a, b]$, we mean a finite set of points $\{x_0, x_1, x_2, \dots, x_n\}$ where $a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$

We write $\Delta x_i = x_i - x_{i-1}$ ($i = 1, 2, 3, \dots$). So Δx_i is the length of the i^{th} sub-interval given by the partition P .

NORM: Norm of a partition P , denoted by $|P| = \max \Delta x_i$. The norm of P is the length of largest subinterval of $[a, b]$ induced by P . Norm of P is also denoted by $\mu(P)$.

REFINEMENT of a Partition: Let P_1 and P_2 be two partitions of $[a, b]$. We say that P_2 is finer than P_1 or P_2 refines P_1 or P_2 is a refinement of P_1 if $P_1 \subset P_2$ i.e. every point of P_1 is a point of P_2 .

Example



UPPER SUMS AND LOWER SUMS

Let f be a bounded function in $[a, b]$

$\therefore$ Necessarily bounded in each sub-interval I_r i.e. $[x_{r-1}, x_r]$

Let M_r, m_r be the bounds in I_r .

Then, Upper sum (Riemann) to the partition P , denoted by $U(P, f)$ defined by

$$U(P, f) = \sum_{r=1}^n M_r \delta_r$$

and lower (Riemann) sum defined by

$$L(P, f) = \sum_{r=1}^n m_r \delta_r$$

where $\delta_r = x_r - x_{r-1}$ (length of r^{th} interval)

Note: For every partition P , $L(P, f) \leq U(P, f)$.



Some Properties of Darboux Sums

- (1) Let f be a bounded function defined on $[a, b]$ and let m, M be the infimum and supremum of f on $[a, b]$. Then for any partition P of $[a, b]$, we have $m(b-a) \leq L(P, f) \leq U(P, f) \leq M(b-a)$
- (2) Let f be a bounded function on $[a, b]$ and let P be a partition of $[a, b]$. If P' is a refinement of P , then
 - (i) $L(P, f) \leq L(P', f)$
 - (ii) $U(P, f) \geq U(P', f)$
 - (iii) $L(P, f) \leq L(P', f) \leq U(P', f) \leq U(P, f)$.
- (3) If P_1 and P_2 are any two partitions of $[a, b]$. Then $U(P_2, f) \geq L(P_1, f)$ i.e. No lower Darboux can exceed any Darboux upper sum.
- (4) If f is a bounded function defined on $[a, b]$ and P be any partition of $[a, b]$, then $U(P, -f) = -L(P, f)$ and $L(P, -f) = -U(P, f)$.

UPPER AND LOWER RIEMANN INTEGRALS

Upper Integral: The infimum of the set of the upper sums $U(P, f)$ is called the upper integral of f over $[a, b]$ and is denoted by

$$\int_a^b f(x) dx$$

The supremum of the set of the lower sums, $L(P, f)$ is called the lower integral of f over $[a, b]$ and is denoted by

$$\int_a^b f(x) dx$$

Thus, $\int_a^b f(x) dx = \inf\{U(P, f) : P \text{ is a partition of } (a, b)\}.$

$$\int_a^b f(x) dx = \sup\{L(P, f) : P \text{ is a partition of } (a, b)\}$$

A bounded function f is said to be Riemann integrable, or simply integrable over $[a, b]$, if its upper and lower integrals are equal; the common value being called the Riemann integral or simply the integral denoted by the symbol

$$\int_a^b f(x) dx$$

Note: The family (or class) of all bounded functions which are Riemann integrable on $[a, b]$ is denoted by $R[a, b]$.

Darboux Theorem

Let f be a bounded function defined on $[a, b]$. Then to every $\epsilon > 0$, there corresponds $\delta > 0$, such that

$$(i) \quad U(P, f) < \int_a^b f(x) dx + \epsilon$$

$$(ii) \quad L(P, f) > \int_a^b f(x) dx - \epsilon$$

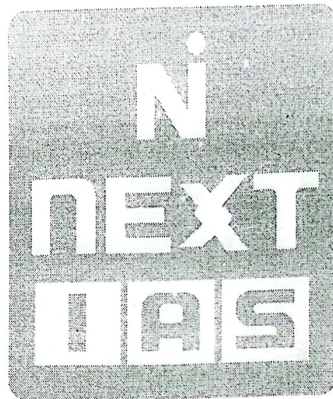


Note: If f is bounded and P is partition of $[a, b]$, then

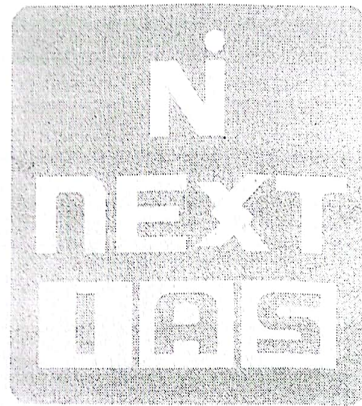
$$(i) \lim_{\|P\| \rightarrow 0} U(P, f) = \int_a^b f(x) dx$$

$$(ii) \lim_{\|P\| \rightarrow 0} L(P, f) = \int_a^b f(x) dx$$

Q. Show that a constant function is integrable and $\int_a^b k dx = k(b-a)$.



Q. Show that $f(x) = \sin x$ is integrable on $[0, \pi/2]$ and $\int_0^{\pi/2} \sin x dx = 1$



Another Definition of Integrability

Consider

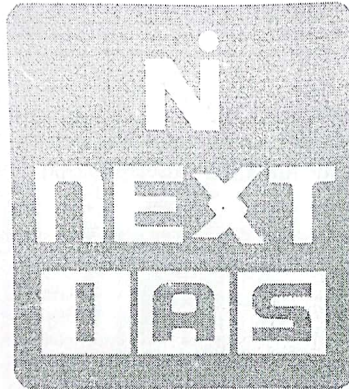
$S(P, f) = \sum_{r=1}^n f(\xi_r)(x_r - x_{r-1}) = \sum_{r=1}^n f(\xi_r)\delta_r$, where $x_{r-1} \leq \xi_r \leq x_r$, then $S(P, f)$ is called a Riemann sum of f over $[a, b]$ with respect to partition P of $[a, b]$.

$$\text{and } \int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} S(P, f) = \lim_{n \rightarrow \infty} S(P, f)$$

or, In less precise manner, we can say:

A function f is integrable, if $\lim_{\|P\| \rightarrow 0} \sum_{r=1}^n f(\xi_r)\delta_r$ exists and is independent of the choice of the interval I_r and of the point ξ_r of I_r ; the limit I , if it exists is called the integral of f over $[a, b]$.

Q. Compute $\int_{-1}^1 f(x) dx$, where $f(x) = |x|$

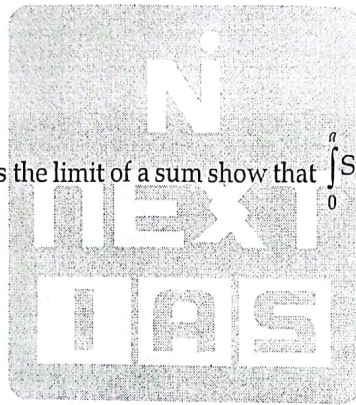


Summation of series

- i. If f is riemann interagble on $[a, b]$, $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=1}^n hf(a + rh)$ where, $h = \frac{b-a}{n}$
- ii. If f is integrable on $[0, 1]$, then $\int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$
- iii. If f is integrate on $[a, b]$, then $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{r=1}^n (ah^r - ah^{r-1}) f(ah^r)$ where $h = \left(\frac{b}{a}\right)^{\frac{1}{n}}$

Q. $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n^2}{(n+1)^3} + \dots + \frac{1}{8n} \right] = \frac{3}{8}$

Q. Using the definition of integral as the limit of a sum show that $\int_0^a \sin x \, dx = 1 - \cos a$

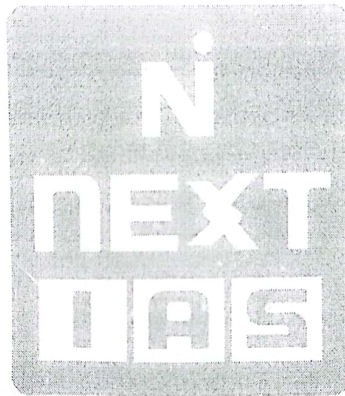


Necessary and sufficient condition for integrability

A NSC for the integrability of a bounded function f is that to every $\epsilon > 0$ there corresponds a partition P such that the corresponding oscillatory sum $\omega(P, f) < \epsilon$ i.e. $U(P, f) - L(P, f) < \epsilon$.

Theorems

1. Every continuous function is integrable.
2. A bounded function f is integrable in $[a, b]$, if the set of its points of discontinuity is finite.
3. A bounded function f is integrable in $[a, b]$ if the set of its points of discontinuity has finite no. of limit points.
4. If f is monotonic in $[a, b]$, then it is integrable in $[a, b]$.



Q. Show that the function f defined as follows:

$$f(x) = \frac{1}{2^n} \quad \text{when } \frac{1}{2^{n+1}} < x \leq \frac{1}{2^n} : n = 0, 1, 2, \dots$$
$$f(0) = 0$$

is integrable in $[0, 1]$, although it has infinite no. of points of discontinuity.



Notable Theorems/Properties

1. If f is bounded and integrable in $[a, b]$, then $|f|$ is also bounded and integrable in $[a, b]$ and $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$.

Converse may not be true,

e.g. $f(x) = \begin{cases} 1 & \text{When } x \text{ is rational} \\ -1 & \text{When } x \text{ is irrational} \end{cases}$

2. If f is integrable on $[a, b]$, then f^2 is also integrable.



Inequalities

1. $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a), \text{ if } b \geq a$

2. If f is bounded and integrable in $[a, b]$ then there exists a number μ , lying between the bounds of ' f ' such that

$$\int_a^b f(x) dx = \mu(b-a) \quad (\text{first mean value Theorem})$$

i.e. $\int_a^b f(x) dx = f(c)(b-a) \text{ where } a \leq c \leq b$

3. Let $|f(x)| \leq K$,

$$\left| \int_a^b f(x) dx \right| \leq K|b-a|$$

4. $f(x) \geq 0 \forall x \in [a, b]$

then $\int_a^b f(x) dx \geq 0$.

5. Let $f \geq g$

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx$$

6. Second Mean Value theorem (Generalised Mean Value theorem)

If $\int_a^b f(x) dx$ and $\int_a^b \phi(x) dx$ both exist and $f(x)$ keeps the same sign, positive or negative throughout the interval of integration, then there exists a number μ , lying between the bounds of f such that

$$\int_a^b f(x)\phi(x) dx = \mu \int_a^b \phi(x) dx$$

'or'

$$= f(\xi) \int_a^b \phi(x) dx \quad (\text{If } f \text{ is continuous also})$$

7. Second Mean Value : (Weierstrass form) If $\int_a^b f(x) dx$ and $\int_a^b \phi(x) dx$ both exist and ϕ is monotonic in $[a, b]$, then there exist $\xi \in [a, b]$ such that

$$\int_a^b f(x)\phi(x) dx = \phi(a) \int_a^{\xi} f(x) dx + \phi(b) \int_{\xi}^b f(x) dx.$$

Bonnet form

If $\int_a^b f(x)dx$ and $\int_a^b g(x)dx$ both exist, g is monotonically decreasing and positive in $[a, b]$, then there exist a point

$\xi \in [a, b]$ then there exist a point such that $\int_a^b f(x)g(x)dx = g(a)\int_a^{\xi} f(x)dx$.

Q. Show that $\frac{1}{\pi} \leq \int_0^1 \frac{\sin \pi x}{1+x^2} dx \leq \frac{2}{\pi}$



Q. If a function f is continuous in $[0, 1]$, show that $\lim_{n \rightarrow \infty} \int_0^1 \frac{nf(x)}{1+n^2x^2} dx = \frac{\pi}{2} f(0)$.

IMPROPER INTEGRAL



Convergence of Improper Integral of First kind

Consider $\int_a^b f(x) dx$

- Convergence at the left end : 'a' is the only point of infinite discontinuity in $[a, b]$

$$\phi(\epsilon) = \int_{a+\epsilon}^b f(x) dx, \quad 0 < \epsilon \leq b-a$$

- Convergence at the right end : 'b' be the only point of infinite discontinuity $\phi(\epsilon) = \int_a^{b-\epsilon} f(x) dx, \quad 0 < \epsilon < b-a$
- Convergence at both the ends : Let a and b be the only points of infinite discontinuity of 'f'.

i. $\int_a^c f(x) dx$

ii. $\int_c^b f(x) dx$

Comparison of two Integrals

Let f and ϕ be two functions such that $\forall x \in [a, b] \quad f(x) \geq 0, \phi(x) \geq 0$. Also let $f(x) \leq \phi(x)$. Then,

i. $\int_a^b \phi(x) dx$ converges $\rightarrow \int_a^b f(x) dx$ converges.

ii. $\int_a^b f(x) dx$ does not converge $\rightarrow \int_a^b \phi(x) dx$ does not converge.

Practical comparison test

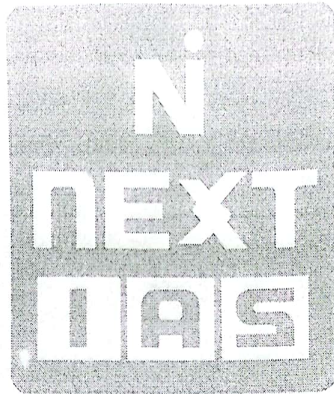
If $\frac{f(x)}{g(x)} \rightarrow l$ when $x \rightarrow a$ and $l \neq 0, \infty$ then the two integrals $\int_a^b f(x) dx$ and $\int_a^b \phi(x) dx$ either both converge or both do not converge.

Note: $\int_a^b \frac{dx}{(x-a)^n}$ converges iff $n < 1$

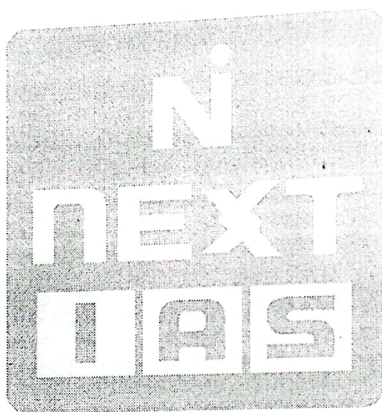
Absolute and Conditional Convergence

Absolute convergent if $\int_a^b |f(x)| dx$ is convergent. Every absolutely convergent integral is convergent.

Q. Show that the Beta function $\int_0^1 x^{m-1}(1-x)^{n-1} dx$ converges iff $m > 0$ and $n > 0$.



Q. Test the convergence of $\int_0^1 \frac{\sin\left(\frac{1}{x}\right)}{\sqrt{x}} dx$



Convergence of improper Integrals of second kind

(Infinite range of integration)

Convergence at ∞

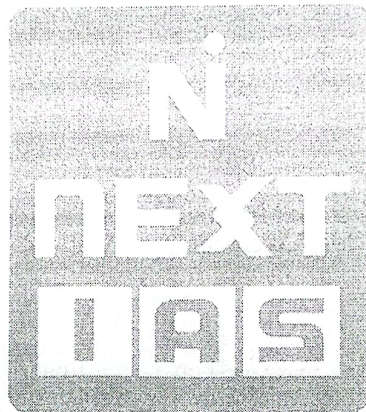
$$\int_a^{\infty} f(x) dx$$

Then $\phi(t) = \int_a^t f(x) dx$, if ϕ tends to a finite limit as $t \rightarrow \infty$, we say that the improper integral is convergent.

Convergence at $-\infty$

If $\int_t^b f(x) dx$ tends to finite limit at $t \rightarrow -\infty$, we say that $\int_{-\infty}^a f(x) dx$ converges.

Q. $\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^2}$



Tests and results for convergence at ∞

1. **Comparison test :** If f and ϕ are two functions such that $0 < f(x) \leq \phi(x) \forall x \in [a, \infty[$, then,

i. $\int_a^\infty \phi(x) dx$ is convergent $\Rightarrow \int_a^\infty f(x) dx$ is convergent

ii. $\int_a^\infty f(x) dx$ is divergent $\Rightarrow \int_a^\infty \phi(x) dx$ is divergent.

2. If f and ϕ are positive on $[a, \infty[$ and $\frac{f(x)}{\phi(x)} \rightarrow l$ when $x \rightarrow \infty$ then,

i. If l is non zero finite, then the two integrals

$$\int_a^\infty f(x) dx \text{ and } \int_a^\infty \phi(x) dx \text{ converge together}$$

ii. If $l = 0$ and $\int_a^\infty \phi(x) dx$ convergent then $\int_a^\infty f(x) dx$ convergent.

iii. If $l = \infty$ and $\int_a^\infty \phi(x) dx$ divergent then $\int_a^\infty f(x) dx$ divergent.

3. $\int_a^\infty \frac{dx}{x^n} (a > 0)$ converges iff $n > 1$

4. Every absolute convergent is convergent.

Q. Show that the Gamma function $\int_0^{\infty} x^{n-1} e^{-x} dx$ is convergent iff $n > 0$



ABEL Test : $\int_a^{\infty} f(x)\phi(x)dx$

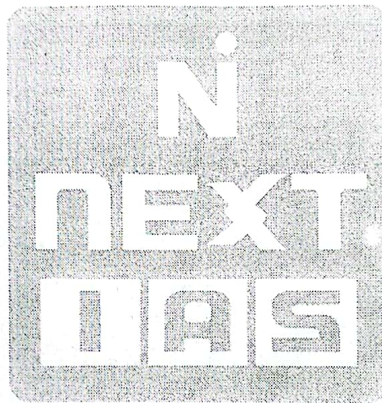
$\Rightarrow \int_a^{\infty} f(x) \rightarrow$ Convergent at ∞ & $\phi(x) \rightarrow$ Bounded and monotonic for $x \geq a$

Then $\int_a^{\infty} f(x)\phi(x)$ converges.

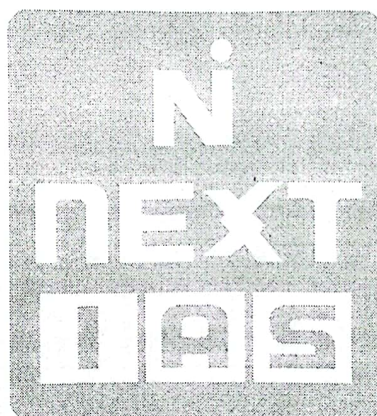
Dirichlet's Test: $\int_a^{\infty} f(x)\phi(x)dx$

$\Rightarrow \int_a^t f(x)dx$ be bounded $\forall t \geq a$ & ϕ be bounded and monotonic in $[a, \infty[$ and $\phi(x) \rightarrow 0$ as $x \rightarrow \infty$

Then $\int_a^{\infty} f(x)\phi(x)$ converges.



Q. $\int_n^\infty \sin x^2 dx$, test for convergence.



Q. Show that $\int_a^{\infty} \frac{\sin x}{x} dx$ converges but not absolutely.

